

YURONG CHEN

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RESEARCH INTERESTS

My research develops theoretical foundations for reliable learning in deployed systems operating in strategic and responsive environments, at the interface of machine learning, algorithmic game theory, and economic design. I study how agents shape the information learners receive and infer learners' objectives, and how endogenous feedback affects learnability and stability. My broader agenda is to design and empirically evaluate learning rules and mechanisms with joint guarantees for learnability, incentives, and welfare when both sides adapt.

ACADEMIC APPOINTMENTS

Inria, École Normale Supérieure, PSL Research University Paris, France

SIERRA team, Inria Paris

Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellow

May 2026–present

Advisors: Francis Bach and Michael I. Jordan

Postdoctoral Researcher

Oct. 2024–Apr. 2026

Advisor: Michael I. Jordan

Research Visits

The University of Hong Kong

Hong Kong, China

Host: Zhiyi Huang

Online prediction and calibrating [C11]

Aug. 2024–Sept. 2024

Learning optimal contracts [C7]

Feb. 2023–Aug. 2023

EDUCATION

Peking University

Beijing, China

Center on Frontiers of Computing Studies (CFCS)

Sept. 2019–Jul. 2024

Ph.D. in Computer Science

Supervisor: Xiaotie Deng

Thesis: *Games over Learning Agents: Private Information Learning, Utilization, and Coordination*

Beihang University

Beijing, China

Hua Luogeng Class, Shen Yuan Honors College

Sept. 2015–Jul. 2019

B.Sc. in Applied Mathematics

GPA: 3.85/4.00 (Top 5%)

Supervisor: Xiangyu Liang

PUBLICATIONS

$(\alpha - \beta)$ denotes alphabetical order (standard in theory venues). * denotes equal contribution.

[†] Industry collaboration (Alibaba: [C8], [C4]; Huawei: [C6], [C2]).

Conference Proceedings

[C11] **Calibrating Made Simple**

$(\alpha - \beta)$ **Yurong Chen**, Zhiyi Huang, Michael I. Jordan, and Haipeng Luo
39th Annual Conference on Learning Theory (**COLT 2026**)

[C10] **Mechanism Design for LLM Fine-tuning with Multiple Reward Models**

Haoran Sun, **Yurong Chen**, Siwei Wang, Chu Xu, Wei Chen, and Xiaotie Deng
39th Conference on Neural Information Processing Systems (**NeurIPS 2025**)

[C9] **Learning a Stackelberg Leader's Incentives from Optimal Commitments**

$(\alpha - \beta)$ **Yurong Chen**, Xiaotie Deng, Jiarui Gan, and Yuhao Li
26th ACM Conference on Economics and Computation (**ACM EC 2025**)

[C8] **Beyond Advertising: Mechanism Design for Platform-Wide Marketing Service**
“**QuanZhanTui**”[†]

Ningyuan Li, Zhilin Zhang, Tianyan Long, Yuyao Liu, Rongquan Bai, **Yurong Chen**, Xiaotie Deng, Pengjie Wang, Chuan Yu, Jian Xu, and Bo Zheng
31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)

- [C7] **Are Bounded Contracts Learnable and Approximately Optimal?**
($\alpha - \beta$) **Yurong Chen**, Zhaohua Chen, Xiaotie Deng, and Zhiyi Huang
25th ACM Conference on Economics and Computation (**ACM EC 2024**)
- [C6] **Coordinated Dynamic Bidding in Repeated Second-Price Auctions with Budgets[†]**
Yurong Chen*, Qian Wang*, Zhijian Duan, Haoran Sun, Zhaohua Chen, Xiang Yan, and Xiaotie Deng
40th International Conference on Machine Learning (**ICML 2023**)
- [C5] **A Scalable Neural Network for DSIC Affine Maximizer Auction Design**
Zhijian Duan, Haoran Sun, **Yurong Chen**, and Xiaotie Deng
37th Conference on Neural Information Processing Systems (**NeurIPS 2023**) – **Spotlight**
- [C4] **Learning-Based Ad Auction Design with Externalities: The Framework and A Matching-Based Approach[†]**
Ningyuan Li, Yunxuan Ma, Yang Zhao, Zhijian Duan, **Yurong Chen**, Zhilin Zhang, Jian Xu, Bo Zheng, and Xiaotie Deng
29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C3] **Optimal Private Payoff Manipulation against Commitment in Extensive-form Games**
($\alpha - \beta$) **Yurong Chen**, Xiaotie Deng, and Yuhao Li
18th Conference on Web and Internet Economics (**WINE 2022**) – **Best Student Paper Award**
- [C2] **On the Convergence of Fictitious Play: A Decomposition Approach[†]**
Yurong Chen, Xiaotie Deng, David Mguni, Chenchen Li, Jun Wang, Xiang Yan, and Yaodong Yang
31st International Joint Conference on Artificial Intelligence (**IJCAI-ECAI 2022**)
- [C1] **The Tight Bound for Pure Price of Anarchy in an Extended Miner’s Dilemma Game**
Qian Wang and **Yurong Chen**
20th International Conference on Autonomous Agents and Multiagent Systems (**AAMAS 2021**)

Journals

- [J1] **Optimal Private Payoff Manipulation against Commitment in Extensive-form Games**
($\alpha - \beta$) **Yurong Chen**, Xiaotie Deng, and Yuhao Li
Games and Economic Behavior, 2025. Journal version of [C3].

WORKING PAPERS

- [P1] **How Sampling Shapes LLM Alignment: From One-Shot Optima to Iterative Dynamics**
($\alpha - \beta$) **Yurong Chen**, Yu He, Michael I. Jordan, and Fan Yao
Selected for **FORC 2026 Highlights Track**.

FELLOWSHIPS AND GRANTS

- **Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship (LIAISON)** 2026–2028
Algorithmic Contract Design for AI Markets: Learning, Information Design, and Robustness to Strategic Manipulation
Independently conceived and wrote the successful proposal.
European Commission; EU contribution: EUR 226,421; 2025-call success rate: 9.6%.
- *Conference travel support*: ACM EC Student Travel Support (2023, 2024)

HONORS AND AWARDS

- **Honorable Mention (1 of 4), China Outstanding Doctoral Dissertation Award in Intelligent Agents and Multi-Agent Systems** – China Computer Federation (CCF) 2025
- **Doctoral Dissertation Incentive Program in Theoretical Computer Science** – CCF 2025
- **Outstanding Doctoral Dissertation Award**, School of Computer Science, Peking University 2024
- **Outstanding Graduate of Peking University** 2024
- **Best Student Paper Award**, 18th Conference on Web and Internet Economics (WINE) – [C3] 2022
- **Excellent Student of CFCS**, Peking University 2019, 2022
- **Merit Student**, Peking University 2021
- **Hongsheng Award**, Peking University 2021
- **Excellent Graduate Student of Beijing** 2019
- **Hua Luogeng Prize**, Chinese Academy of Sciences 2016–2018
- **National Scholarship** (departmental top 5%) 2016, 2017
- **Excellent Student**, Beihang University 2016, 2017
- **Academic Excellence Award**, Beihang University, 1st Prize 2016, 2017
- **Award for Subject and Competition**, Beihang University, 2nd Prize 2017
- **National English Competition for Undergraduate Students**, 2nd Prize 2017

- **China Undergraduate Mathematical Contest in Modeling**, Beijing District, 2nd Prize 2017
- **National Mathematics Competition for Undergraduate Students** (math major), 2nd Prize 2016

INDUSTRY RESEARCH AND ENGAGEMENT

Alibaba Group – Alimama, Ad Product Technology

Research collaboration and mentorship, including an Algorithm Engineer internship (May–Sept. 2024)

Proposed the position-based externality model in [C4], yielding a maximum-weight bipartite-matching implementation of the Score-Weighted VCG framework. Mentored junior researchers across two mechanism-design projects; both KDD papers ([C4], 2023; [C8], 2025) use real-world and synthetic data.

Huawei – Research Collaborations

Led the research in both collaborations: proved decomposition-based fictitious-play convergence conditions [C2]; formulated coordinated bidding for budget-constrained advertisers and co-developed pacing algorithms with asymptotic per-advertiser utility guarantees relative to independent bidding [C6].

Academic–Industry Convening: Co-organizer, AIMS Workshop at ICLR 2026, alongside researchers from Alibaba Group and academia, with sponsorship from Alimama.

TEACHING

Guest Lecturer

University of North Carolina at Chapel Hill

- **COMP 790-191: Game Theory in Learning** (graduate course) Jan. 2026
Learning a Stackelberg Leader’s Incentives from Optimal Commitments

Teaching Assistant

Peking University

- **Conscious Turing Machine** Spring 2020, 2021, 2023
- **Mathematics in Artificial Intelligence**, Yuanpei College (selective, interdisciplinary) Spring 2022
- **Discrete Mathematics I**, Turing Class (selective undergraduate CS program) Fall 2019, 2020
- **Discrete Mathematics II**, Turing Class Spring 2020

Beihang University

- **Real Analysis**, Hua Luogeng Class, Shen Yuan Honors College Spring 2019

Responsibilities: Led recitation sessions; designed assignments and exams; contributed to course materials reused in subsequent offerings; held office hours.

PROFESSIONAL SERVICE

Guest Editor

Theoretical Computer Science – Special Issue of COCOA 2024

Track Chair

Conscious AI track, IJTCS-FAW 2022 & 2023; Conscious Turing Machine track, IJTCS 2021

Workshop Organizer

Gender Inclusion Workshop (GIW) at EC 2026; AI for Mechanism Design and Strategic Decision Making (AIMS) Workshop at ICLR 2026

Mathematical Foundations of the Internet and Blockchain Economics Workshop (student organizer, 2024 & 2025)

Program Committee Member

ACM EC 2025 & 2026, WWW 2026, WINE 2025 & 2026, IJTCS-FAW 2025

Conference Reviewer

SODA 2025 & 2027, FOCS 2024 & 2026, ICML 2026 (Silver Reviewer Award), AISTATS 2025 & 2026, ESA 2025, NeurIPS 2024 & 2025, STACS 2025, ICLR 2025, ITCS 2025, ISAAC 2023, STOC 2023, WINE 2020

Journal Reviewer

Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), Journal of Combinatorial Optimization (JOCO), Theoretical Computer Science (TCS), SIAM Journal on Computing (SICOMP)

Volunteer

Conferences: EC 2023, IJTCS-FAW 2022 & 2023, WINE 2021, IJTCS 2021

Forum: The 2nd Cybereconomy and Game Theory Forum in China

SELECTED TALKS

Calibrating Made Simple

- 39th Annual Conference on Learning Theory (COLT 2026), San Diego, CA, USA, Jul. 2026
- Easy Peasy Workshop at ACM EC 2026, Rome, Italy, Jul. 2026
- JECCO UK 2026, All Souls College, University of Oxford, Oxford, UK, Jul. 2026

Learning a Stackelberg Leader's Incentives from Optimal Commitments

- **Keynote**, Online Learning and Economics (OLE) Workshop at ACM EC 2026, Rome, Italy, Jul. 2026
- FairPlay Seminar, Criteo, Paris, France, Apr. 2026
- Max Planck Institute for Intelligent Systems, Tübingen, Germany, Nov. 2025
- INFORMS Annual Meeting, Atlanta, GA, USA, Oct. 2025
- King's College London, London, UK, Sept. 2025
- London School of Economics, London, UK, Sept. 2025
- Imperial College Business School, London, UK, Sept. 2025
- Foundations of AI Seminar, University of Warwick, Coventry, UK, Sept. 2025

How Sampling Shapes LLM Alignment: From One-Shot Optima to Iterative Dynamics

- 2026 ESIF Economics and AI+ML Meeting, Cornell University, Ithaca, NY, USA, Jun. 2026
- EconCS Talk, Harvard University, Cambridge, MA, USA, Jun. 2026
- FORC 2026 Highlights Track, Harvard University, Cambridge, MA, USA, Jun. 2026
- National University of Singapore, Singapore, Mar. 2026
- Nanyang Technological University, Singapore, Mar. 2026
- The University of Hong Kong, Hong Kong, China, Mar. 2026
- CSE Seminar, The Chinese University of Hong Kong, Hong Kong, China, Mar. 2026
- CIS Seminar, University of Macau, Macau, China, Mar. 2026
- CSE Seminar, The Hong Kong University of Science and Technology, Hong Kong, China, Mar. 2026
- CS Colloquium, City University of Hong Kong, Hong Kong, China, Feb. 2026

Are Bounded Contracts Learnable and Approximately Optimal?

- University of Oxford, Oxford, UK, Jun. 2025
- FOAM Seminar, University of Amsterdam, Amsterdam, Netherlands, Mar. 2025
- 15th Day on Computational Game Theory, Tilburg, Netherlands, Mar. 2025
- INFORMS Annual Meeting, Seattle, WA, USA, Oct. 2024
- City University of Hong Kong, Hong Kong, China, Aug. 2024

Optimal Private Payoff Manipulation against Commitment in Extensive-form Games

- GAMES 2024, the 7th World Congress of the Game Theory Society, Beijing, China, Aug. 2024
- 2023 IJCAI Young Elite Symposium (IJCAI-Young 2023), Shanghai, China, Jul. 2023

Coordinated Dynamic Bidding in Repeated Second-Price Auctions with Budgets

- Frontiers of Online Advertising: Autobidding, GenAI, and Beyond Workshop at ACM EC 2024, New Haven, CT, USA, Jul. 2024
- Female Forum, IJTCS-FAW 2023, Macau, China, Aug. 2023